WORKED SOLUTION

STEP III 2015, Question 7

Pure • Differential Operators

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q7

The key idea

Powers of x are eigenfunctions of $\mathrm{D} = x\frac{\mathrm{d}}{\mathrm{d}x}$: applying D to x^a just multiplies it by a. So D^n acts on a polynomial term by term, multiplying the coefficient of x^r by r^n, which is exactly where the r^n in the final sum comes from. For the divisibility, notice that each application of D can lower the power of a factor $(1-x)$ by at most one.

Solution

The opening result

For any constant a (and $x > 0$ if a is not an integer),

$$\mathrm{D}x^a = x \cdot ax^{a-1} = ax^a.$$

Applying D again and using this result with the constant a pulled out,

$$\mathrm{D}^2x^a = \mathrm{D}(ax^a) = a\,\mathrm{D}x^a = a^2x^a,$$

as required. Since D is linear ($\mathrm{D}(\alpha f + \beta g) = \alpha\,\mathrm{D}f + \beta\,\mathrm{D}g$ for constants α, β), repeating the step gives, by induction,

$$\mathrm{D}^n x^a = a^n x^a \qquad (n \geq 1).$$

Part (i)

Write $\mathrm{P}(x) = \sum_{k=0}^{r} c_k x^k$ with $c_r \neq 0$. By linearity and the result above,

$$\mathrm{D}^n\mathrm{P}(x) = \sum_{k=0}^{r} c_k\, k^n x^k.$$

This is a polynomial with no terms beyond x^r, and its x^r coefficient is $r^n c_r$. Because $r \geq 1$, we have $r^n \neq 0$, and $c_r \neq 0$, so $r^n c_r \neq 0$. Hence $\mathrm{D}^n\mathrm{P}(x)$ is a polynomial of degree exactly r.

(The condition $r \geq 1$ matters: a constant polynomial is sent to 0, because $0^n = 0$.)

Part (ii)

Fix m. We prove by induction on n that for $1 \le n \le m$,

$$\mathrm{D}^n(1-x)^m = (1-x)^{m-n}\,\mathrm{Q}_n(x) \quad \text{for some polynomial } \mathrm{Q}_n.$$

Base case. $\mathrm{D}(1-x)^m = x\cdot(-m(1-x)^{m-1}) = (1-x)^{m-1}\cdot(-mx)$, so $\mathrm{Q}_1(x) = -mx$.

Inductive step. Suppose the statement holds for some n with $n < m$, so that $m - n \ge 1$. By the product rule,

$$\begin{aligned}\mathrm{D}^{n+1}(1-x)^m &= x\frac{\mathrm{d}}{\mathrm{d}x}\Big[(1-x)^{m-n}\mathrm{Q}_n(x)\Big]\\ &= x\Big[-(m-n)(1-x)^{m-n-1}\mathrm{Q}_n(x) + (1-x)^{m-n}\mathrm{Q}_n'(x)\Big]\\ &= (1-x)^{m-n-1}\cdot x\Big[(1-x)\mathrm{Q}_n'(x) - (m-n)\mathrm{Q}_n(x)\Big].\end{aligned}$$

The bracket is a polynomial, so this has the required form with $\mathrm{Q}_{n+1}(x) = x[(1-x)\mathrm{Q}_n'(x) - (m-n)\mathrm{Q}_n(x)]$.

> **?** **Ask yourself** Where did we use $n < m$ in the inductive step?

Taking out the factor $(1-x)^{m-n-1}$ only makes sense as a polynomial factor if $m - n - 1 \ge 0$, that is, $n < m$. At $n = m$ there is no factor of $(1-x)$ left to lose, so the induction (correctly) stops there.

So for every positive integer $n < m$, $\mathrm{D}^n(1-x)^m$ is divisible by $(1-x)^{m-n}$.

Part (iii)

Expand by the binomial theorem:

$$(1-x)^m = \sum_{r=0}^{m}(-1)^r\binom{m}{r}x^r.$$

Applying D^n term by term, exactly as in part (i),

$$\mathrm{D}^n(1-x)^m = \sum_{r=0}^{m}(-1)^r\binom{m}{r}r^n x^r.$$

By part (ii), with $n < m$, the left-hand side equals $(1-x)^{m-n}\mathrm{Q}_n(x)$ with $m - n \ge 1$, so it vanishes at $x = 1$. Setting $x = 1$ on the right-hand side therefore gives

$$\sum_{r=0}^{m}(-1)^r\binom{m}{r}r^n = 0,$$

as required. (The $r = 0$ term is $0^n = 0$, consistent with D killing the constant term.)

Checks

Take $m = 3$, $n = 2$: the sum is $0 - 3 \cdot 1 + 3 \cdot 4 - 1 \cdot 9 = 0$. Take $m = 4$, $n = 3$: $0 - 4 + 6 \cdot 8 - 4 \cdot 27 + 64 = 0$. The identity was also confirmed by computer for all $1 \le n < m \le 19$, together with the fact that $(1-x)^{m-n}$ is the *exact* power dividing $\mathrm{D}^n(1-x)^m$ for $m \le 11$. As a contrast, when $n = m$ the sum is no longer zero: for $m = 2$ it is $0 - 2 + 4 = 2 = 2!$, and in general it is $(-1)^m m!$.

Common slips

- Writing $\mathrm{D}^2 x^a$ as $x^2 \frac{\mathrm{d}^2}{\mathrm{d}x^2} x^a = a(a-1)x^a$: D^2 is D applied twice, not x^2 times the second derivative.
- In part (i), forgetting to say why the leading coefficient $r^n c_r$ is non-zero; this is where $r \ge 1$ is used.
- In part (iii), substituting $x = 1$ into $(1-x)^m$ *before* applying D^n; the point is that the factor $(1-x)$ survives $n < m$ applications.

Beyond STEP

The operator $x\frac{\mathrm{d}}{\mathrm{d}x}$ is sometimes called the Euler operator, and x^a is an eigenfunction with eigenvalue a; it is the natural tool for Euler–Cauchy equations such as $x^2y'' + bxy' + cy = 0$, which become constant-coefficient equations under $x = e^t$. The identity in part (iii) says that the mth finite difference of a polynomial of degree less than m is zero, and it is closely linked to Stirling numbers and inclusion–exclusion counts of surjections. Boyce and DiPrima, *Elementary Differential Equations*, covers Euler equations; Siklos, *Advanced Problems in Mathematics*, has more problems in this style.

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